

The artifact is a text-based Hangman game written in C++ where a player guesses a secret word letter by letter with the word either entered by another player or randomly pulled from a text file by the computer. It was originally created as a CIS 125 final project and is split across three files. It has a header, an implementation file, and a main driver which follows standard C++ structure. I chose this artifact because it gave me a clear opportunity to show growth beyond just making something that works. The biggest improvements were breaking the original single bloated class into four focused classes, renaming vague variables to something descriptive, cutting out repeated logic by moving it into helper functions, and adding input validation so the program handles bad input gracefully instead of breaking.

I feel like I met the outcomes I set out to hit. The refactoring shows I can apply real software engineering principles like separation of concerns and the defensive programming additions show I was thinking about edge cases and reliability. One thing I'd add going forward is planning class responsibilities on paper before writing any code because doing it after the fact is a lot harder than building it right the first time. The biggest thing I took away from this was learning the difference between code that works and code that's actually well designed. The hardest part was deciding where one class's responsibility ended, and another's began especially when it felt easier to just leave things where they were. Cleaning up the duplicate guess checking logic was also a good reminder that when you are solving the same problem in two places it usually means something wasn't organized clearly to begin with.